

Bench	Task		Trials	Raw	+Workflow	+Skills	Raw Tok	WF Tok	SK Tok	ΔP_{SK-Raw}	ρ_{SK-Raw}
Terminal-Bench-2											
TB2	model-extraction-relu-logits		5	60.0%	20.0%	0.0%	482K	109K	185K	-60.0%	-61.7%
TB2	crack-7z-hash		5	80.0%	0.0%	40.0%	986K	441K	576K	-40.0%	-41.6%
TB2	overfull-hbox		5	80.0%	60.0%	40.0%	555K	446K	415K	-40.0%	-25.2%
TB2	torch-tensor-parallelism		5	40.0%	0.0%	0.0%	174K	147K	170K	-40.0%	-2.3%
TB2	financial-document-processor		5	60.0%	0.0%	20.0%	1.1M	1.0M	1.5M	-40.0%	+41.2%
TB2	tune-mjcf		5	20.0%	0.0%	0.0%	453K	215K	0K	-20.0%	-100.0%
TB2	adaptive-rejection-sampler		5	20.0%	80.0%	20.0%	1.7M	579K	2.2M	+0.0%	+31.8%
TB2	chess-best-move		5	80.0%	100.0%	80.0%	361K	164K	314K	+0.0%	-12.9%
TB2	path-tracing		5	80.0%	0.0%	80.0%	2.2M	369K	2.1M	+0.0%	-0.9%
TB2	sanitize-git-repo		5	80.0%	100.0%	80.0%	2.2M	1.0M	1.0M	+0.0%	-53.4%
TB2	build-pov-ray		5	80.0%	100.0%	100.0%	1.4M	1.4M	1.3M	+20.0%	-6.6%
TB2	fix-code-vulnerability		5	80.0%	80.0%	100.0%	1.4M	1.1M	614K	+20.0%	-56.5%
TB2	large-scale-text-editing		5	80.0%	100.0%	100.0%	403K	288K	422K	+20.0%	+4.9%
TB2	log-summary-date-ranges		5	80.0%	100.0%	100.0%	163K	171K	248K	+20.0%	+52.4%
TB2	mcmc-sampling-stan		5	80.0%	20.0%	100.0%	10.8M	492K	4.3M	+20.0%	-59.9%
TB2	path-tracing-reverse		5	80.0%	0.0%	100.0%	4.0M	872K	3.3M	+20.0%	-16.7%
TB2	portfolio-optimization		5	80.0%	100.0%	100.0%	391K	289K	307K	+20.0%	-21.4%
TB2	query-optimize		5	40.0%	0.0%	60.0%	642K	303K	603K	+20.0%	-6.1%
TB2	regex-chess		5	80.0%	60.0%	100.0%	3.0M	964K	3.1M	+20.0%	+1.6%
TB2	rstan-to-pystan		5	80.0%	0.0%	100.0%	4.1M	1.1M	4.9M	+20.0%	+18.8%
TB2	winning-avg-corewars		5	80.0%	60.0%	100.0%	4.7M	545K	1.4M	+20.0%	-69.1%
TB2	qemu-startup		5	20.0%	60.0%	40.0%	758K	362K	137K	+20.0%	-81.9%
TB2	hf-model-inference		5	60.0%	20.0%	80.0%	1.1M	446K	355K	+20.0%	-68.3%
TB2	mteb-leaderboard		5	60.0%	0.0%	80.0%	19.8M	1.6M	8.4M	+20.0%	-57.5%
TB2	dna-insert		5	20.0%	40.0%	60.0%	454K	276K	394K	+40.0%	-13.2%
TB2	pytorch-model-recovery		5	40.0%	60.0%	80.0%	794K	191K	634K	+40.0%	-20.2%
TB2	write-compressor		5	60.0%	20.0%	100.0%	248K	64K	592K	+40.0%	+138.3%
TB2	break-filter-js-from-html		5	40.0%	40.0%	100.0%	376K	255K	749K	+60.0%	+99.1%
TB2	cancel-async-tasks		5	40.0%	0.0%	100.0%	69K	99K	168K	+60.0%	+143.5%
TB2	extract-elf		5	40.0%	0.0%	100.0%	722K	221K	216K	+60.0%	-70.1%
TB2	pypi-server		5	20.0%	20.0%	100.0%	499K	273K	378K	+80.0%	-24.3%
SkillsBench											
SB	weighted-gdp-calc		5	80.0%	80.0%	0.0%	633K	450K	318K	-80.0%	-49.7%
SB	protein-expression-analysis		5	80.0%	100.0%	20.0%	529K	283K	343K	-60.0%	-35.2%
SB	data-to-d3		5	60.0%	40.0%	20.0%	882K	691K	621K	-40.0%	-29.5%
SB	paper-anonymizer		5	40.0%	20.0%	20.0%	940K	878K	769K	-20.0%	-18.2%
SB	trend-anomaly-causal-inference		5	20.0%	20.0%	0.0%	1.9M	7.5M	2.4M	-20.0%	+25.1%
SB	earthquake-phase-association		5	80.0%	60.0%	80.0%	2.1M	1.6M	1.3M	+0.0%	-41.7%
SB	energy-market-pricing		5	60.0%	40.0%	60.0%	866K	711K	665K	+0.0%	-23.2%
SB	fix-druid-loophole-cve		5	20.0%	20.0%	20.0%	13.4M	9.8M	17.2M	+0.0%	+28.3%
SB	fix-erlang-ssh-cve		5	60.0%	100.0%	60.0%	2.3M	1.6M	1.2M	+0.0%	-47.4%
SB	grid-dispatch-operator		5	40.0%	20.0%	40.0%	635K	393K	400K	+0.0%	-36.9%
SB	hvac-control		5	80.0%	60.0%	80.0%	220K	354K	266K	+0.0%	+21.0%
SB	dapt-intrusion-detection		5	80.0%	80.0%	100.0%	277K	244K	310K	+20.0%	+11.7%
SB	glm-lake-mendota		5	80.0%	80.0%	100.0%	2.0M	1.9M	1.3M	+20.0%	-35.7%
SB	adaptive-cruise-control		5	20.0%	20.0%	40.0%	5.7M	3.7M	3.7M	+20.0%	-34.9%
SB	organize-messy-files		5	20.0%	40.0%	40.0%	602K	666K	577K	+20.0%	-4.1%
SB	fix-visual-stability		5	60.0%	100.0%	80.0%	4.9M	3.3M	4.5M	+20.0%	-7.3%
SB	invoice-fraud-detection		5	60.0%	0.0%	80.0%	359K	319K	241K	+20.0%	-32.7%
SB	r2r-mpc-control		5	60.0%	60.0%	80.0%	526K	352K	354K	+20.0%	-32.8%
SB	react-performance-debugging		5	40.0%	100.0%	80.0%	3.8M	2.6M	4.3M	+40.0%	+11.9%
SB	video-filler-word-remover		5	40.0%	100.0%	80.0%	2.8M	1.9M	562K	+40.0%	-79.7%
SB	dialogue-parser		5	40.0%	40.0%	100.0%	488K	438K	452K	+60.0%	-7.4%
SB	financial-modeling-qa		5	40.0%	40.0%	100.0%	827K	776K	698K	+60.0%	-15.6%
SB	virtualhome-agent-planning		5	40.0%	80.0%	100.0%	284K	187K	259K	+60.0%	-8.9%
SB	gravitational-wave-detection		5	20.0%	60.0%	100.0%	2.5M	1.3M	593K	+80.0%	-76.7%
Terminal-Bench-Pro											
TB-Pro	build-uci-chess-engine-to-beat-multiple-bots		5	40.0%	40.0%	0.0%	1.1M	817K	742K	-40.0%	-33.0%
TB-Pro	decode-go-ctf-credentials		5	80.0%	60.0%	40.0%	958K	799K	539K	-40.0%	-43.7%
TB-Pro	fix-xor-neural-network-instability		5	80.0%	80.0%	60.0%	544K	571K	424K	-20.0%	-22.1%
TB-Pro	reverse-engineer-kvstore-binary-protocol		5	80.0%	100.0%	60.0%	810K	412K	869K	-20.0%	+7.3%
TB-Pro	implement-chemical-equilibrium-solver		5	20.0%	0.0%	0.0%	215K	280K	282K	-20.0%	+30.7%
TB-Pro	portfolio-risk-analyzer-script		5	20.0%	20.0%	0.0%	182K	149K	212K	-20.0%	+16.2%
TB-Pro	mongodb-sales-aggregation-engine		5	60.0%	40.0%	60.0%	167K	136K	133K	+0.0%	-20.5%
TB-Pro	fix-gomoku-win-detection		5	80.0%	100.0%	100.0%	92K	85K	112K	+20.0%	+21.6%
TB-Pro	improve-code-similarity-feature-extraction		5	80.0%	100.0%	100.0%	803K	1.3M	491K	+20.0%	-38.8%
TB-Pro	secure-postgres-schema-migration		5	80.0%	60.0%	100.0%	634K	387K	479K	+20.0%	-24.5%
TB-Pro	train-loan-default-logreg-model		5	80.0%	100.0%	100.0%	147K	82K	159K	+20.0%	+8.5%
TB-Pro	implement-depgraph-dependency-resolver		5	20.0%	20.0%	40.0%	468K	212K	220K	+20.0%	-53.0%
TB-Pro	implement-nonogram-puzzle-solver		5	60.0%	60.0%	80.0%	130K	220K	248K	+20.0%	+90.5%
TB-Pro	optimize-urdf-robot-for-pybullet		5	60.0%	80.0%	80.0%	1.6M	1.2M	645K	+20.0%	-59.5%
TB-Pro	train-matrix-factorization-embeddings		5	60.0%	80.0%	80.0%	351K	173K	270K	+20.0%	-23.0%
TB-Pro	train-sarsa-taxi-agent		5	60.0%	100.0%	80.0%	422K	516K	292K	+20.0%	-30.6%
TB-Pro	advanced-json-to-rfc4180-csv-converter		5	60.0%	80.0%	100.0%	401K	426K	548K	+40.0%	+36.6%
TB-Pro	compare-cmake-build-types-performance		5	60.0%	100.0%	100.0%	111K	89K	145K	+40.0%	+30.8%
TB-Pro	flood-escape-path-planner		5	60.0%	80.0%	100.0%	214K	241K	303K	+40.0%	+41.5%
TB-Pro	present-known-plaintext-key-recovery		5	60.0%	100.0%	100.0%	5.9M	907K	1.2M	+40.0%	-79.9%
TB-Pro	python-jacobi-eigen-solver-conversion		5	60.0%	80.0%	100.0%	198K	237K	185K	+40.0%	-6.2%
TB-Pro	python-sql-injection-sanitizer		5	40.0%	100.0%	80.0%	201K	156K	181K	+40.0%	-9.7%
TB-Pro	repair-broken-shell-data-pipeline		5	60.0%	100.0%	100.0%	759K	623K	850K	+40.0%	+12.0%
TB-Pro	setup-mlflow-sqlite-server		5	60.0%	60.0%	100.0%	1.0M	789K	348K	+40.0%	-66.2%
TB-Pro	text-image-ocr-pipeline		5	60.0%	100.0%	100.0%	109K	128K	145K	+40.0%	+33.1%
TB-Pro	validate-and-solve-sudoku		5	60.0%	100.0%	100.0%	438K	338K	326K	+40.0%	-25.7%
TB-Pro	implement-tensor-parallel-matmul		5	40.0%	100.0%	100.0%	165K	419K	177K	+60.0%	+7.5%
TB-Pro	jq-github-contributor-report		5	40.0%	40.0%	100.0%	111K	139K	219K	+60.0%	+98.3%
TB-Pro	normalize-invoice-pdfs-to-csv		5	40.0%	20.0%	100.0%	212K	135K	234K	+60.0%	+10.2%
TB-Pro	regex-bitcoin-p2pkh-extraction		5	40.0%	100.0%	100.0%	65K	73K	111K	+60.0%	+70.9%
TB-Pro	train-fraud-detection-model		5	40.0%	60.0%	100.0%	180K	235K	163K	+60.0%	-9.6%
TB-Pro	recover-and-sanitize-git-db-credential		5	20.0%	80.0%	80.0%	548K	418K	491K	+60.0%	-10.4%
TB-Pro	sokoban-bfs-move-with-fallback		5	20.0%	80.0%	80.0%	111K	166K	202K	+60.0%	+81.8%